William Graves

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SUMMARY

Electrical engineering junior (graduating May 2028) focused on semiconductor processing, thin films, and process equipment, seeking a Summer 2027 internship in process, equipment, or integration engineering. Hands-on experience with DOE-driven process studies, statistical analysis in JMP, thin-film metrology, and defect characterization, backed by process simulation, equipment design, and data acquisition.

EDUCATION

Arizona State University Expected May 2028

Bachelor of Science in Engineering, Electrical Engineering GPA: 4.00

Relevant Coursework: Properties of Electronic Materials (quantum theory of solids, carrier transport, pn junctions; studied abroad in Germany), Fundamentals of Solid-State Devices (MOS capacitors, MOSFETs, BJTs, metal-semiconductor contacts), Fundamentals of CMOS & MEMS Processing, Circuits I/II, Signals & Systems, Electromagnetics

TECHNICAL SKILLS

Process & Metrology: Spin coating, photolithography, thermal oxidation, cleanroom protocol, Dektak XT profilometry, Gwyddion, Filmetrics, spectroscopic ellipsometry (Woollam M-2000), optical microscopy, defect inspection, curve-tracer I-V

Data Analysis & Process Control: Design of Experiments (DOE), SPC, JMP/JSL, Python, MATLAB, C++, ImageJ/Fiji, PostgreSQL/SQL

CAD, Simulation & Hardware: Silvaco Athena, LTspice, SolidWorks, Fusion 360 (CAD/CAM), KiCad, Arduino/ESP32 DAQ, Ignition SCADA, oscilloscopes

RESEARCH EXPERIENCE

Undergraduate Researcher - TSMC-Sponsored FURI Program Fall 2026

School of Electrical, Computer and Energy Engineering; advisor Prof. Michael Kozicki

- Characterizing 405 nm SLA resin as a low-cost photoresist on Si: film intact to 1550 rpm, lifted off above 2100 rpm; cured thickness 6.1-7.3 µm at 1000 rpm to 0.8 µm at 2000 rpm, fit with a rinse-depth model in JMP.
- Measured thickness by Bruker Dektak XT stylus profilometry with Gwyddion leveling after ellipsometry failed on the thick films.
- Designed a 2^2 full-factorial DOE with 3 center points (ramp time, final speed; dwell fixed at 60 s) for dewetting-defect onset; analysis in JMP.
- Built an automated time-lapse microscope imaging station (Python/OpenCV, Arduino Opta relay for illumination) and a Fiji/ImageJ macro that counts defects in a fixed central ROI.

Undergraduate Researcher - ECRAM Process Integration Fall 2026-Present

School of Electrical, Computer and Energy Engineering; Prof. Matthew Marinella's group

- Beginning development of a CMOS-integrated, back-end-of-line process flow for three-terminal ECRAM analog memory for AI hardware; in the process of getting trained in the ASU NanoFab toward fabricating the devices.

PROJECTS

Process Simulation & Metrology - EEE 435, ASU NanoFab Fall 2026

- Gowned into the ASU NanoFab cleanroom; trained in contamination control and cleanroom protocol.
- Fit the Deal-Grove model to 1050 °C wet-oxidation data (B = 0.415 µm²/hr, A = 0.272 µm, within 2% of measured past 5 min), and compared oxide and LPCVD nitride uniformity from Filmetrics and M-2000 ellipsometry data.
- Simulated Si fin oxidation in Athena: thins up to 41% faster than planar Deal-Grove; oxidation/HF trim from 400 to 200 nm.
- Modeled GaAs RIE with an oxide hard mask in Athena: extracted etch rates and selectivity (6.9:1), then ran a DOE to find the shortest etch (9.3 min, 0.17 µm oxide) for a 1 µm trench.
- Measured bird's-beak encroachment in poly-buffered LOCOS: 1.85 µm of a 2.2 µm drawn active region stays usable.

Device I-V Characterization - EEE 334

- Measured Zener diode (1N5234/37/39) turn-on and breakdown voltages on a curve tracer, within 6% of rated values.
- Extracted CD4007 NMOS Vt, kn'(W/L), λ, ro, and gm, and checked a common-source amplifier against LTspice.

Equipment Team Lead - Sun Devil Semiconductor Club Fall 2025-Present

- Lead the club's process and characterization equipment. Designed and built a programmable wafer spin coater: CNC-machined aluminum chuck (Fusion 360 CAD/CAM), closed-loop Arduino motor control, and a custom operator interface; data logging via Ignition SCADA and PostgreSQL.

Battery Management Subteam - Sun Devil Motorsports EV Fall 2025-Spring 2026

- Designed an ESP32-based temperature DAQ PCB in KiCad, multiplexing NTC thermistors for up to 304 sense points on a ~1,200-cell FSAE accumulator; supported CAN communication and high-voltage accumulator integration.

CERTIFICATIONS & AWARDS

Design for Six Sigma (DFSS) Yellow Belt, 2026 | Eagle Scout, Boy Scouts of America, 2022